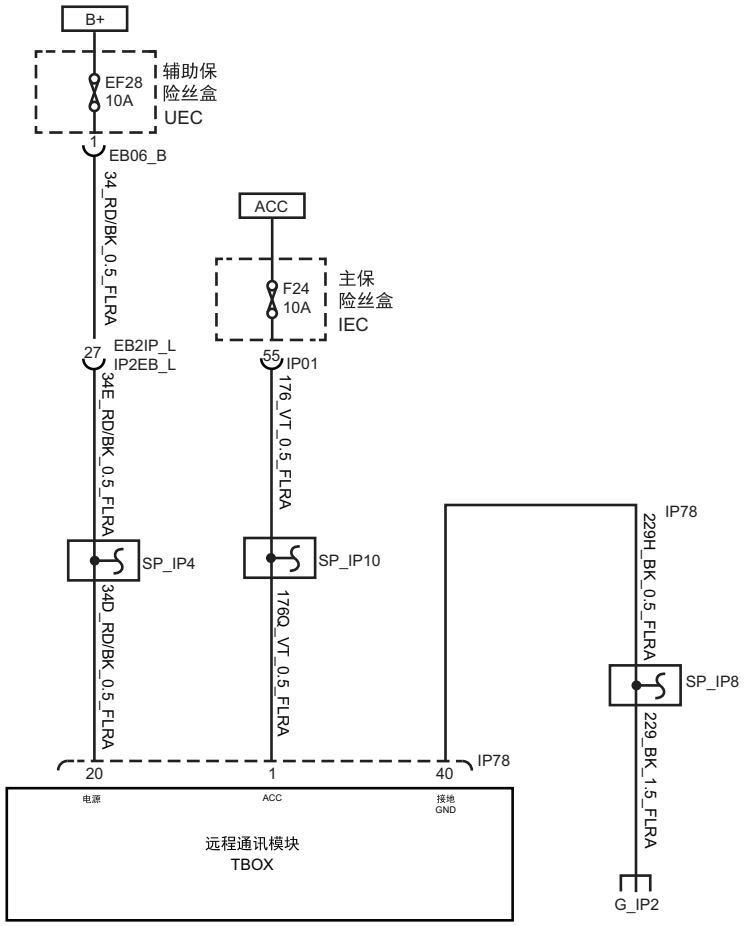
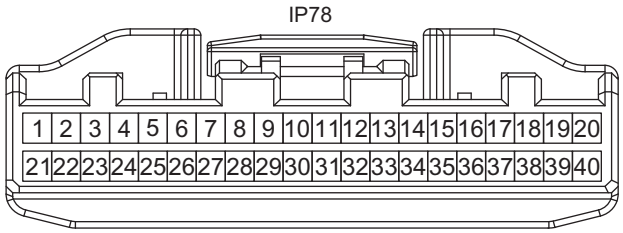
B1541-00、 B1542-00、 U1500-62、 U1562-17、 U1563-16、 B1570-31

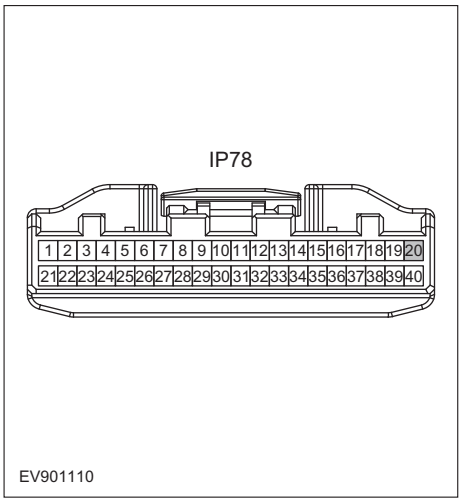
Description of fault diagnosis codes
B1541-00：TBOX(Telematics Box) Internal SIM Card Status - Not Present
B1542-00：TBOX(Telematics Box) Internal SIM Card - Invalid
U1500-62：VIN Mismatch with BCM(Body Control Management) (or IPK(Instrument Park))
U1562-17：Battery voltage high
U1563-16：Battery voltage low
TB1570-31：BOX(Telematics Box) Bluetooth module fault
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Battery • TBOX • Charging system • Insurance fuse
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1608-1C
B1609-1C
To set the effect of a fault code condition
Description of circuit diagram
The TBOX will monitor whether all sensors and actuators are within normal range all the time. Meanwhile,it will check the connector for virtual connection, short circuit, open circuit and the connection rationality with other signals. When fault occurs, the TBOX module will save the fault code corresponding to that fault and enable safety mode.

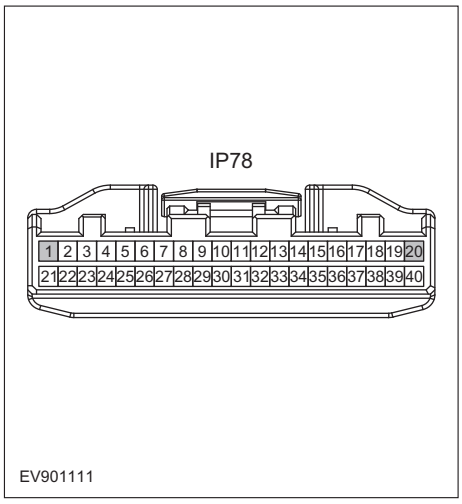
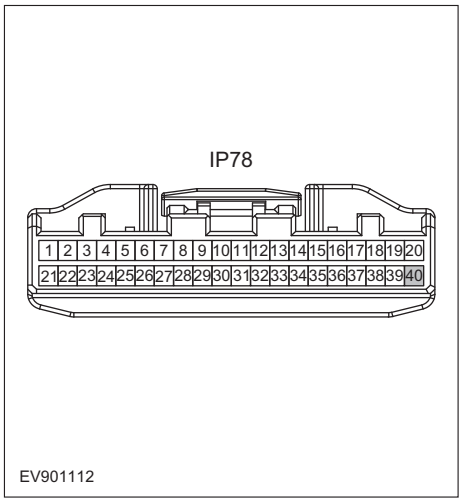
Circuit diagram





EV901109

Troubleshooting steps		
Step 1	General inspection.	
		1 Check fuses EF28/F24 for damage. 2 Check battery capacity. 3 Check the harness plug of TBOX for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the TBOX with diagnostic apparatus. 4 If there is any fault code except for B1541-00、B1542-00、U1500-62、 U1562-17、 U1563-16、 B1570-31?
		YES → Refer to:DTC Summary List (TBOX).
NO ↓		
Step 3	Check the power voltage of TBOX battery.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP78 of TBOX. 3 Measure the voltage between the harness terminal of TBOX and ground. Measuring circuit: voltage between terminal 20 on plug IP78 and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the power voltage when the TBOX is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of TBOX and ground. Measuring circuit: voltage between terminal 1 on plug IP78 and ground. Measuring circuit: voltage between terminal 20 on plug IP78 and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Test whether the grounding circuit of TBOX is open.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP78 of TBOX. 3 Measure the resistance between the harness terminal of TBOX and ground. Measuring circuit: resistance between terminal 40 on plug IP78 and ground. **Standard value: < 1Ω** 4 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Test whether TBOX is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug IP78 of TBOX. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the TBOX plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the TBOX.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	